rived in various parts of the world. Just that, they carry things instead of people. Autonomous delivery systems are being deployed in environments ranging from mines, war fields, and hospitals to warehouses, industries, airports, college campuses, and residential localities.

We have already touched upon their use in the industrial space in our story on cobots (collaborative robots). We have seen how they work alongside human workers, delivering spare parts, samples and many other goods, when and where required. In this story, let us explore the potential of autonomous last-mile delivery, which promises to be the near-future precursor to self-driving cars.

Why the bot

The need for autonomous last-mile delivery was felt badly during the pandemic period when most merchants were open only for delivery but it was difficult to find people for the job. Customers were also favouring contactless deliveries. This gave autonomous delivery vehicles and bots a much-needed push.

However, Scott Rosenberger, Global Transportation, Hospitality & Services sector leader, Deloitte, states in a report that, "While driverless delivery technology has gathered momentum during the pandemic, it will easily outlast this period with both companies and consumers recognising and, already, reaping benefits from it." Here are some reasons why the technology picked up and is likely to continue growing in the coming years:

• During the pandemic, there was a dire need to deliver goods, especially food, safely to people. Customers feared human delivery agents, who could be carriers of disease, and preferred to deal with these little bots instead. Autonomous delivery services often use contactless authentication—such as scanning of a QR code sent to the customer's phone—to open the box locker and retrieve their goods, without punching any keys.

• Most AVs are electric-powered and often utilise renewable power sources for charging. Autonomous delivery fleets are usually controlled by smart, cloud based software systems, which help to optimise routes and reduce energy consumption further. All of this helps reduce emissions and meet sustainability goals.

• Autonomous delivery systems help overcome the problem of inaccessibility. According to a Deloitte report, even before the pandemic, remote regions of the world were using drone delivery and ziplines for essential supplies. This system has further developed to deliver a wider variety of goods.

• Also, since AVs or robots generally have a smaller form factor with shorter wheelbase, and dynamic map support to beat the city traffic, they are able to reach places faster and better even within the city.

• Since they move as per software-suggested routes, and can be tracked all the time, the delivery time is also more predictable and accurate.

• Ford and Volkswagen closed down their self-driving technology initiative Argo AI as they felt that creating a robo-taxi capable of navigating in a dense urban landscape is harder than putting a man on the moon. Part of the challenge is technological, while part of it is regulatory. In most regions, government regulations prohibit fast, self-driving cars on public roads. And their concern is quite understandable. However, most regulatory bodies are prepared to allow, or at least consider, autonomous last-mile delivery systems as these are quite small, carry smaller payloads, weigh less, and it would not hurt much even if it inadvertently bumped into a human. A small cart obviously looks less lethal than a huge truck or car without anyone behind the wheels. Direct delivery devices are permitted in several American cities, including Chicago, Detroit, and Pittsburgh. It is deemed as a way to reduce vehicular congestion and cut down on emissions. Several places in the UK, China, the Middle East, and Africa also permit autonomous deliveries.

• Autonomous delivery systems do away with the uncertainties related to manual labour, such as availability and pricing.

• The technological world has made great strides in sensing, mapping, artificial intelligence (AI), and cloud based deployments. And there is this new-born realisation that it is better to put self-driving technologies to use in a smaller format for more immediate use-cases. All this has given autonomous delivery the fillip it deserves.

According to a Fortune Business Insights study, the global autonomous last-mile delivery market is projected to grow from $11.12 billion in 2021 to $51.38 billion by 2028 at a compound annual growth rate (CAGR) of 24.4% in the forecast period. Allied Market Research projects that the market will reach approximately $90 billion by 2030. Another report pins it at approximately $80 billion. Values vary but most market research firms are making positive predictions for autonomous last-mile delivery.

There are several modes of autonomous last-mile delivery—drones, ziplines, small vehicles, or robots and trucks. Going forward, deployments are likely to be multi-modal, with trucks for moving goods from the warehouses to retailers, small vehicles and robots to deliver to the consumers' houses, and drones and ziplines for hard-to-reach destinations across rough ter-